

Backpaper - Optimization (2023-24)

Time: 3 hours.

Attempt all questions. The total marks is 50.

1. Write down the dual of the following problem: minimize $x_1 - x_2$

$$\begin{aligned} \text{subject to} \quad & 2x_1 + 3x_2 - x_3 + x_4 \leq 0, \\ & 3x_1 + x_2 + 4x_3 - 2x_4 \geq 3, \\ & -x_1 - x_2 + 2x_3 + x_4 = 6, \\ & x_1 \leq 0, \\ & x_2, x_3 \geq 0. \end{aligned} \quad [4 \text{ points}]$$

2. A polyhedron is represented by a system of equality and inequality constraints.

- (a) Give an example of a polyhedron P and a point $\mathbf{x}$, and two representations of P , such that $\mathbf{x}$ is a basic solution under one representation but is not a basic solution under the other representation. [2 marks]
- (b) Can one create an example of P with two representations, and an $\mathbf{x}$ such that $\mathbf{x}$ is a basic *feasible* solution (bfs) under one representation and not a bfs under the other representation? Explain. [2 marks]
3. Consider the following polyhedron in standard form: $2x_1 + x_2 + 2x_3 + x_4 = 3$, $x_1 + x_2 + 5x_3 + 3x_4 = 2$ and all $x_1, x_2, x_3, x_4 \geq 0$.
- (a) Find the basic feasible solution where x_1, x_2 are the basic variables. [2 mark]
- (b) Find the 3rd basic direction. [2 marks]
- (c) If we are minimizing $x_1 + x_2 + x_3 + x_4$ over the polyhedron, what is the reduced cost along the 3rd basic direction? [2 marks]

4. Let $f : \mathbf{R}^n \rightarrow \mathbf{R}$ be convex and differentiable. Show that $\mathbf{x}^*$ solves

$$\min_{\mathbf{x} \in \mathbf{R}^n} f(\mathbf{x})$$

if and only if $\nabla f(\mathbf{x}^*) = \mathbf{0}$. [6 marks]

5. Describe the *gradient descent algorithm* for minimizing a convex differentiable function $f : \mathbf{R}^n \rightarrow \mathbf{R}$. [4 marks]
6. Consider the constrained optimization problem

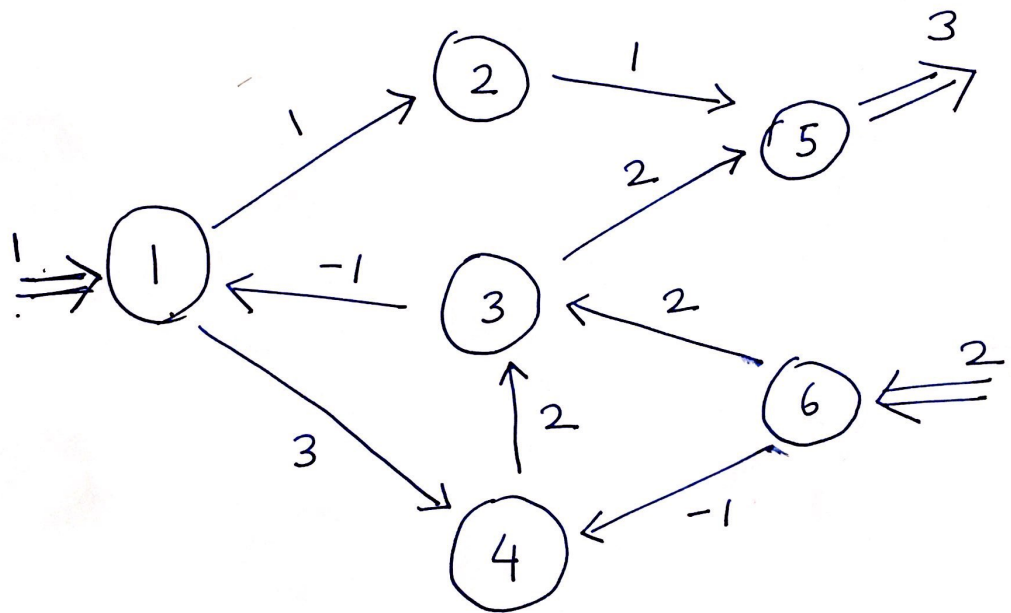
$$\min_{\mathbf{x} \in \mathbf{R}^n} f(\mathbf{x}) \quad \text{subject to } g_m(\mathbf{x}) \leq 0, \quad m = 1, 2, \dots, M, \quad (0.1)$$

where $f, g_1, \dots, g_M : \mathbf{R}^n \rightarrow \mathbf{R}$. Corresponding to the above problem, for each choice of real numbers $\lambda_1, \dots, \lambda_M$, is an unconstrained problem

$$\min_{\mathbf{x} \in \mathbf{R}^n} \left[f(\mathbf{x}) + \sum_{m=1}^M \lambda_m g_m(\mathbf{x}) \right]$$

- (a) Write down the dual problem. [2 marks]
- (b) Let $\boldsymbol{\lambda}^*$ be the solution of the dual problem, and $\mathbf{x}^*$ be the solution of the primal problem (0.1). Show that if strong duality holds, then $\boldsymbol{\lambda}^*$ is exactly what is needed to make $\mathbf{x}^*$ the solution to the unconstrained problem. [6 marks]
7. Minimize $-2x_1 + x_2 - 2x_3$

$$\begin{aligned} \text{subject to} \quad & 2x_1 + x_2 + x_4 = 10, \\ & x_1 + 2x_2 - 2x_3 + x_5 = 20, \\ & x_2 + 2x_3 + x_6 = 5, \\ & x_1, x_2, \dots, x_6 \geq 0. \end{aligned} \quad [8 \text{ points}]$$



8. Consider the *uncapacitated* network flow problem on the directed graph above. The numbers next to each directed arc $\rightarrow$ is the *cost* associated to the arc, while the numbers next to $\Rightarrow$ is the external *supply/demand* at the node.

Denote by $\mathbf{c}$ the vector of costs corresponding to the arc. We are interested in minimizing the total cost $\mathbf{c}^T \mathbf{f}$, where the flow vector $\mathbf{f}$ satisfies the flow conservation equations and $\mathbf{f} \geq \mathbf{0}$.

- Find an optimal basic feasible solution (feasible tree solution) to the problem.
(You can start with the tree given by the dotted lines) [8 marks]
- Find the optimal cost for the problem. [2 marks]